

Variables:

- Variable is a named memory location
- Variables are used to store values
- We can store values by using Assignment Operator (=)
- Assignment Operator is also called as Right to Left operator

Syntax: Variablename = Value
Left Right

Ex: age = 22
pcost = 256.26
name = 'AVD'

age → Variable name
22 → Value

Syntax2: Variablename = Value / Operand / Expression

Ex: a = 5
b = a
c = p --> Error, Because we don't have any variable with 'p'

c = a + b

Note: In One Variable we can store only one value at a time, whenever second value comes then first value is deleted

Ex: a = 5
a = 7
print(a)

Output: 7

Ex: a = 7
b = 5
a = a + b
b = a - b
print(a)
print(b)

Output: 12
7

Ex: a = 3
b = 5
c = a
a = b
b = c
print(a)
print(b)

Output: 5
3

Ex:

```
p = 5
q = 7
p = p + q
q = p - q
p = p - q
print(p)  --> 7
print(q)  --> 5
```

p	q
5	7
12	5
7	5

--> We can declare more than one variable at a time

Syntax: Variable1, Variable2, = Value1, Value2,

No. of Variables and No. of values should be same otherwise we will get Error

Ex: ~~a, b = 3, 5~~

```
print(a) --> 3
print(b) --> 5
```

Ex: ~~p, q, r = 7, 3~~

```
print(p)
print(q)
print(r)
```

-- Error --

--> If you want to develop any program follow 3 steps

1. Collect Values then save it --> Variables
2. Perform Operations --> Operators --> +, -, *, /
3. Display Result --> print()

--> Consider pizzacost is 300.0, Customer order for 3 pizzas now find billAmt

```
{ pizzacost = 300.0
  qty = 3
✓ billAmt = pizzacost * qty
✓ print(billAmt)
```

print() Function

--> print() function is used to display data

--> In print() function inside of single or double quotes whatever we wrote something we will get as an output

Ex:

```
print('Hai')      ---> Hai
print("Hello")    ---> Hello
print('2+3')      ---> 2+3
a = 5
print('a')        ---> a
```

Note: If you want to perform operations then write without using single or double quotes

Ex:

```
print(2+3)        ---> 5
a = 7
print(a)          ---> 7
```

print Output as 'Your result is 7'

print('Your result is ', a) (or) print(f'Your result is {a}')

Display message as It's Great Day

--> Consider firstname as 'AVD' and last name as 'Group' now find fullname

```
firstname = 'AVD'
lastname = 'Group'
fullname = firstname + lastname
print(fullname)
```

--> consider a value is 5, b value is 7.2, now find sum then print sum

<pre>a = 5 b = 7.2 sum = a + b print(sum)</pre>		<pre>a = 5 b = 7.2 sum = a + b print("Sum is : ",sum)</pre>		<pre>a = 5 b = 7.2 sum = a + b print(f'Sum is {sum}')</pre>
---	--	---	--	---

--> Consider Annual salary is 600000, now find monthly salary

```
annualsal = 600000
monthllysal = annualsal/12
print(monthllysal)
```

--> Consider 1kg onions cost is Rs60, customer order for 5 kgs now find billAmt

```
onioncost = 60
qty = 5
billAmt = onioncost * qty
print(billAmt)
```

--> Consider student marks are 95, 90, 94. now find total, avg and print total, avg

```
s1 = 95
s2 = 90
s3 = 94
total = s1 + s2 + s3
avg = total/3
print('Total is : ',total)
print('Average is : ',avg)
```

Reading user input:

- input() function is used to get a value from user
- One input() function can take only one value at a time
- It always takes values in string format only

Syntax: `Variablename = input() / input('Your message ')`

Ex: W.a.p to read ename then print empname

```
ename = input('Enter Emp Name : ')
print('Your Name is : ', ename)
```

Ex2: W.a.p to read fname, lname then find fullname

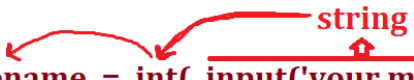
```
fname = input('Enter FirstName : ')
lname = input('Enter LastName : ')
fullname = fname + ' ' + lname
print(fullname)
```

TypeConversion

--> Converting a value from one type to another type is called as typeconversion

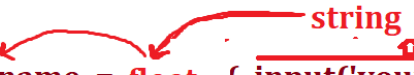
Read integer value :

Syntax: `Variablename = int( input('your message') )`



Read Floating Value :

Syntax: `Variablename = float ( input('your message') )`



Based on user input either int/float

Syntax: `Variablename = eval ( input('your message') )`

Ex: W.a.p to read two integer numbers then find sum

```
a = int( input('Enter any Number : ') )
b = int( input('Enter any Number : ') )
sum = a + b
print(sum)
```

Ex: W.a.p to read 1 integer and 1 floating value then find multiplication

```
a = int(input('Enter any Number : '))
b = float(input('Enter any Number : '))
mul = a * b
print(mul)
```

-->W.a.p to read emp annual salary then find monthly salary

```
annual_sal = float(input('Enter Annual Salary : '))  
monthly_sal = annual_sal/12  
print(f'Monthly Salary is {monthly_sal}')
```

--> Consider pizzacost is 300.0, now read quantity then find bill Amount

```
pizzacost = 300.0  
qty = int(input('Enter Quantity : '))  
billAmt = pizzacost * qty  
print(f'Your Bill Amount is :{billAmt}')
```



```
print('Your Bill Amount is : ', billAmt)
```